Project # 3 Approach:

Which data science skills are most valued?

Composition of the group

S.No	Name of Student	Email
01	Muhammad Imran	Muhammad.imran91@stu-mail.sps.cuny.edu
02	Kenny Jeannite	kenny.jeannite72@cuny.edu
03	Samantha Mesa	samantha.mesa84@login.cuny.edu
04	Tenzin Thakuri	tenzin.thakuri35@login.cuny.edu
05	Shanice Smith	shanice.smith16@cuny.edu

Introduction and Background

After an in-depth discussion of the research question, **"Which data science skills are most valued?"**, the group concluded that addressing this topic requires an empirical, data-driven approach that goes beyond merely listing ubiquitous programming languages or routine, repetitive tasks. To establish a rigorous foundation for this study, it is essential to first define what constitutes "value." In this regard, the group asserts that the value of a data science skill is best measured by its ability to create distinct differentiation within an organization.

Rather than viewing skills through the lens of basic technical proficiency, true value manifests when competencies enable a data science team or individual to drive competitive advantage, solve non-standard business challenges, and propel strategic transformation. Under this framework, organizational differentiation can be operationalized across three key dimensions:

1. **Strategic Impact vs. Commodity Work:** Differentiating skills move professionals away from replaceable execution (e.g., standard baseline modeling or basic reporting) toward

high-leverage capabilities, such as translating complex domain problems into actionable algorithmic frameworks or designing end-to-end architectures.

2. **Problem-Solving Complexity & Novelty:** Value is defined by the capacity to tackle unstructured, ambiguous problems that off-the-shelf automated AI tools cannot resolve. Skills in advanced statistical inference, causal modeling, and custom system design allow organizations to innovate rather than merely keep pace.
3. **Value Creation and Measurable ROI:** Skills create high differentiation when they directly influence enterprise key performance indicators (KPIs)—whether by optimizing core operational workflows, unlocking new revenue streams, or providing proprietary insights competitors cannot easily replicate.

By defining value through organizational differentiation, this project establishes a quantitative framework to evaluate labor market data, job market premiums, and business outcomes, ultimately pinpointing the exact capabilities that elevate data science from a functional utility to a strategic asset.

Potential Data sources to back our view in this project:

GitHub Job Market Repositories: The group agrees to use the results of the 15th Annual Stack Overflow Developer Survey, the longest-running and most comprehensive study of those who code in the industry. For a decade and a half, the insights gathered from responses have profoundly shaped the global developer community's understanding of the technology landscape, influencing tools, careers, and industry at large.

In this dataset, the questions are divided into 5 sets: technology, community, and AI questions are randomized, including learning and career, tech and tech culture, community & AI, and final thoughts.

Proposed Analytical Pipeline

1. Text Parsing & Skill Extraction (NLP): This is step 1; in this step, the data that we have gathered is unstructured text (.csv file). We will need to build a matching dictionary

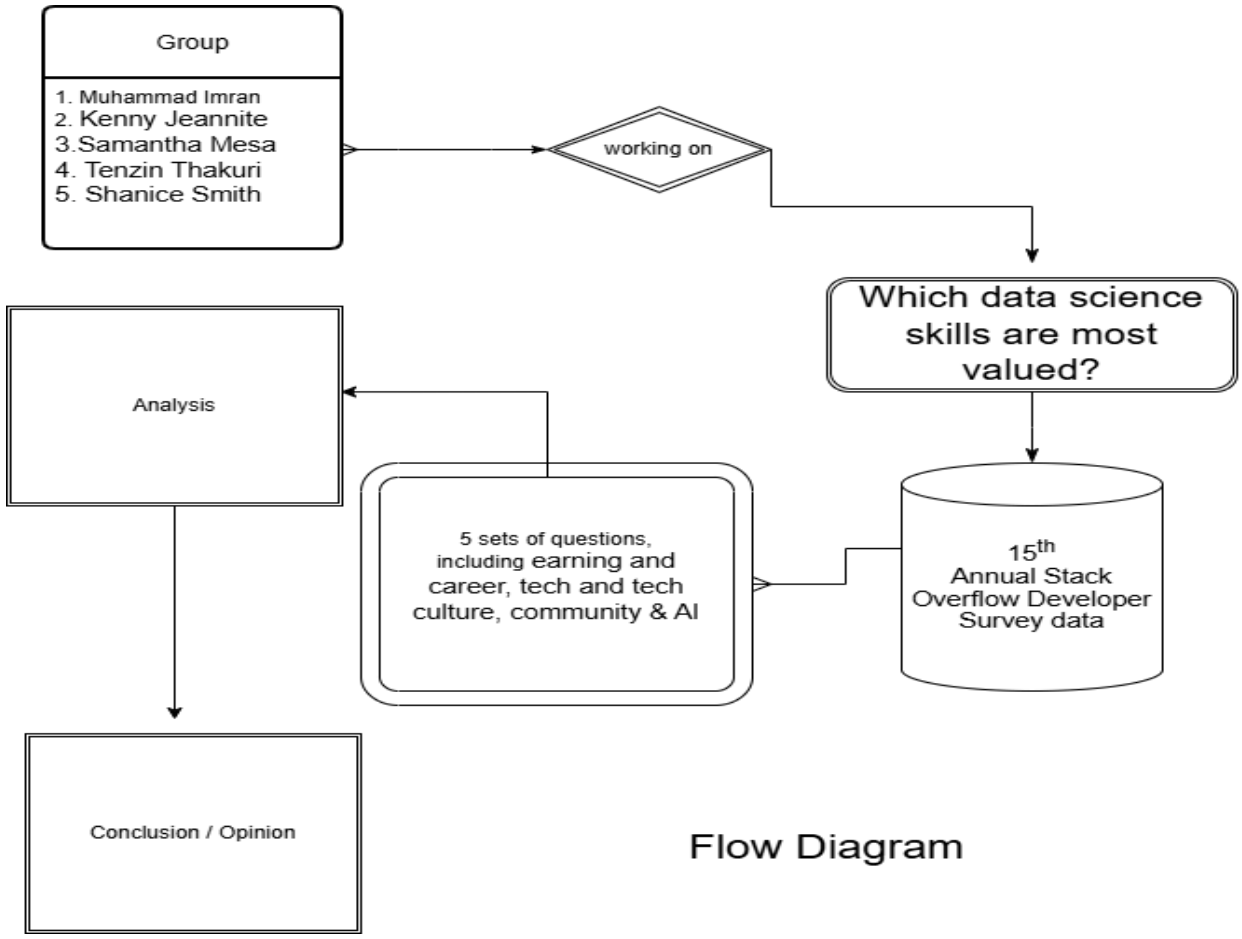
or use regular expressions (regex) and natural language processing (spaCy) to extract standardized skill tokens from raw descriptions.

2. Exploratory Data Analysis & Co-Occurrence: In this step, we are going to substantiate our argument that a skill's value is driven by its ability to create organizational differentiation rather than routine execution, we leverage the provided survey dataset (15th Annual Stack Overflow Developer Survey) in several concrete, empirical ways.

   a. Demonstrating Compensation Premiums for Differentiating Roles (Economic Differentiation): Group respondents by their specific developer roles or advanced skill sets (e.g., cdomparing standard software developers or entry-level execution roles against specialized architects, data/AI engineers, or senior leadership roles) and calculate their median and mean `ConvertedCompYearly`. It will substantiate our argument that if basic coding were the sole driver of value, compensation would scale linearly with years of experience alone. By showing that roles focused on high-level architecture, specialized analytics, or strategic system design command massive salary premiums, we empirically prove that organizations pay exponentially more for capabilities that create strategic differentiation.

   b. Job Satisfaction and Organizational Impact Analysis: Cross-reference `JobSat` (job satisfaction scores) and `ConvertedCompYearly` across respondents who spend their time on routine, repetitive tasks versus those who report handling complex system design, problem-solving, or strategic deployment, it substantiates our argument in a way that data often shows that professionals engaged in low-leverage, repetitive execution face lower career satisfaction and stagnant compensation growth. Conversely, those equipped with differentiating problem-solving skills report higher satisfaction and career leverage, reinforcing that true value creates both individual and organizational distinction.

Data Flow Diagram /ERD



Suggested Project Structure

1. **Introduction & Motivation:** Why the modern data landscape requires a granular look at skill valuation.

2. **Data Collection & Cleaning:** Overview of your dataset size, filtering criteria, and NLP keyword extraction pipeline.

3. **Exploratory Findings:** Visualizations showing top skills by frequency, segmented by job roles (Data Analyst vs. Data Scientist vs. ML Engineer).

4. **Deep Dive – The Premium Skills:** As we get out from the data.

5. **Conclusion & Recommendations:** Actionable takeaways for professionals building a learning roadmap.